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Assumptions and distributions

The values live in `config.py`; change them there and rerun. “Realized” comes from `validate.py` on seed 20261108. Where a public benchmark exists, the parameter was tuned so the realized rate lands near it. Where none was found, the value is my own assumption and is marked **assumption**.

Tessaly Health is fictional and sits in an unnamed region, so it uses **national** averages. No local market and no employer data were used.

Sources

Key	Source	Used for
NSI-26	NSI Nursing Solutions, *2026 National Health Care Retention & RN Staffing Report* (calendar 2025; 527 hospitals)	RN and hospital turnover, first-year turnover, vacancy, time to recruit, cost per turnover
NSI-25	NSI Nursing Solutions, *2025 National Health Care Retention & RN Staffing Report* (calendar 2024; 450 hospitals)	Same measures for 2024, turnover by specialty
ECEC	BLS, Employer Costs for Employee Compensation, RNs in private hospitals, Dec 2024 (TED article, May 2025)	Wage per hour worked and benefits load
OEWS	BLS, Occupational Employment and Wage Statistics, SOC 29-1141, May 2024 and May 2025	Wage growth year to year
SIA	SIA/NATHO Travel Nurse Benchmarking Survey, selected findings 2025 and 2026	Agency bill rates
NRP	Vizient/AACN Nurse Residency Program outcomes, as reported by Vizient (2025)	Residency retention check

1. Company and window

Assumption	Value	Benchmark	Realized
Window	1 Jan 2023 to 31 Dec 2025		
Facilities	1 flagship hospital, 2 community hospitals, 20 clinics, home health, system office	locked decision	25
Units and departments	41 nursing units, 79 non-nursing departments		120
Budgeted positions	4,122	“about 4,000 staff”	3,888 to 3,938 filled
Nursing share	RNs, LPNs, nursing assistants, nurse managers	“roughly half”	49%
RN position vacancy	8%/5%/3% open on day 1 by tier, plus 6%/4%/2% of RN positions that stay open all window (hard to fill)	NSI-26: 8.6% national RN vacancy	8.8–8.9%

Turnover tiers follow NSI’s specialty ranking. **High:** Behavioral Health, Emergency, Step-Down, Medical-Surgical (NSI-26: 22.5%, 20.7%, 19.0%, 18.1%). **Low:** Perioperative, Women and Infants (NSI-26: 14.9%, 15.0%), plus clinics and Nursing Administration. **Medium:** Intensive Care, Float Pool, Home Health.

2. Workforce flow

Assumption	Value	Benchmark	Realized (2023 / 2024 / 2025)
Voluntary turnover, year 2+	RN high 17%, medium 13%, low 10%; LPN/NA 15–22%; clinical support 14%; administrative 12%; facilities/EVS 22%; managers 7%		see next rows
All RNs		NSI-25: 16.4% (2024); NSI-26: 17.6% (2025)	19.1% / 16.4% / 15.4%
High-tier RNs		NSI-26 high specialties: 18.1–22.5%	23.7% / 20.9% / 21.9%
Low-tier RNs		NSI-26 low specialties: 13.4–15.0%	14.6% / 10.1% / 10.9%
All hospital staff		NSI-25: 18.3%; NSI-26: 18.5%	20.1% / 17.8% / 18.3%
First-year multiplier	Year-1 hazard × 2.0	NSI-26: 22.7% of new RNs leave within a year	21.5% (2023 hires), 17.4% (2024 hires)
Involuntary turnover	2% per year	**assumption**	
Time to fill	Gamma(shape 2); mean 95 days high-tier RN, 75 other RN, 45 everyone else	NSI-26: 78 days to recruit an experienced RN (range 56–102)	
Starting tenure	Gamma(shape 1.3, scale 5 years), capped at 35	**assumption**	
FTE	1.0 (62%), 0.9 (14%), 0.75 (10%), 0.6 (9%), 0.5 (5%)	**assumption**	
Internal transfer fills	10% of RN fills, 6% of others	**assumption**	161 transfers
Rehires	5% of external fills	**assumption**	103

Assumption	Value	Benchmark	Realized (2023 / 2024 / 2025)
New-grad RNs	35% of inpatient RN hires, 5% elsewhere	**assumption**	197 residents

Low-tier RN turnover runs a little under NSI's lowest specialties. Clinics and administration are mixed into that tier, and NSI surveys hospitals only.

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3. Planted effects (case study B)

These are the true answers an analysis should try to recover. They're in `ground_truth/planted_effects.csv`.

ID	Planted effect	Built-in trap
E-01	Year-1 voluntary exit hazard × 0.70 if role onboarding is finished within 30 days. Sized to be detectable with about 2,500 hires (**assumption**)	On-time rate is 80% normally, 65% on high-turnover units, which also lose more people. Naive ratio on this seed: 0.74.
E-02	Year-1 hazard × 0.65 for new-grad RNs with attendance propensity of 0.80 or higher (Beta(5, 2), minus 0.10 on high-turnover units) (**assumption**)	Completers must stay 12 months, so they show 0% year-1 exits: immortal-time bias. About 130 residents have a full year of follow-up, with about two dozen first-year exits, so any honest estimate is noisy. Realized resident first-year exit is 20%, above the NRP-reported 11% (89% retention). NRP's figure covers participating programs, so it's likely a favorable comparison; I kept the gap rather than tune to it.
E-04	Days to proficiency: new grad Gamma(mean 140, sd 35) if high attender, (170, 45) if not; experienced (75, 25); agency (12, 4); × 1.2 on ICU, ED, Step-Down (**assumption**)	Nurses who leave first are censored, not missing.

4. Staffing and cost (case study C)

Assumption	Value	Benchmark	Realized
Required RN hours	Budgeted RNs × mean FTE × 36 h/week ÷ 7 × (1 − 7% absence)	**assumption**	
Demand variation	+6% Dec–Feb, −3% Jul–Aug, daily noise sd 4%	**assumption**	
Coverage order	Overtime (25–40% of the gap, capped at 12% of required; 60–80% where no agency), then agency, then unfilled	**assumption**; no public US benchmark found for overtime share	High tier: agency 6.4%, OT 4.4%, unfilled 2.8%
Agency booking	Monthly, 13-week contracts covering 90% of the past 28 days' gap, 14-day lead, 25% extension chance	**assumption**; 13 weeks is the standard travel contract	497 agency nurses, 634 segments
Staff RN wage per hour worked	$49.10 / $51.09 / $53.24 (2023–25)	ECEC: $51.09 in Dec 2024. Other years scaled about 4% a year from OEWS growth (approximate)	
Benefits load on wages	56%	ECEC: benefits $28.43 vs. wages $51.09 per hour worked	Fully loaded staff RN about $79.70 in 2024
Overtime premium	1.5 × wage	federal minimum overtime premium	
Agency bill rate	$106.63 / $90.13 / $90.54	SIA national aggregate	2024 agency premium over a fully loaded staff hour: about 13%
Cost per RN turnover	$61,110 (2024), $60,090 (2025)	NSI-25, NSI-26	

Assumption	Value	Benchmark	Realized
Agency orientation per contract	$1,200	**assumption**	

What the benchmarks change for case C. With public figures, an agency hour costs only about 13% more than a fully loaded staff hour in 2024. That's down from about 39% in 2023. So the case for converting agency nurses rests less on the hourly rate and more on turnover cost, orientation, time to proficiency and continuity. Also, NSI's turnover cost already includes some vacancy coverage, so adding it to agency hours would double-count. The case study has to pick one.

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5. Learning records (case studies A and D)

No public benchmark was found for training completion behavior, so this whole section is **assumption**.

Assumption	Value	Realized
Annual compliance	5 modules, new version each year; assigned 1 Feb, due 30 Apr	
New-hire compliance	Hires from 15 Jan on get current modules at hire, due in 30 days	Hires 15 Jan–1 Feb get both: a natural duplicate
Completion behavior	Base 88% on time, with person (sd 0.6) and unit (sd 0.3) effects on the log-odds, −0.3 on high-turnover units; of the rest, 65% late (due + Exp(21 days)), 35% never; 4% skip one module	Raw on-time 74–77% of enrollments, before cleanup
Assessments	7% fail first try, then retake; 10% of those fail twice	
Exemptions	2.5% leave of absence per cycle; 1% equivalent credit for Workplace Safety	302 and 102
Certifications	Life Support (2 yr), Equipment (1 yr), Specialty Unit (2 yr), Hazardous Materials (1 yr); renewal 60 days before expiry; 85% on time, 10% late, 5% lapse	19,247 records
Onboarding	Orientation due in 14 days, role onboarding in 30, nursing unit checklist in 60	
Residency	Cohorts Feb, Jun, Sep; 12 monthly sessions; complete = 10+ attended and still employed at session 12	197 residents
LMS accounts	Nightly HR feed; 5% of deactivations lag 8–40 days; 30% of rehires get a new account; 2% of staff have a manual duplicate; agency accounts manual, 75% with an agency ID	6,910 accounts

6. Planted data-quality and pipeline issues (case studies A and D)

Listed with row counts in `ground_truth/planted_issues.csv`.

ID	Issue
DQ-01	Some 2025 learners completed the 2024 version of Information Security
DQ-02	Exact duplicate passed-attempt rows
DQ-03	Enrollments marked completed with no attempt
DQ-04	Due date before assigned date
DQ-05	2024 Fire and Emergency campaign reloaded at Community North; duplicates never completed
DQ-06	2025 Respectful Workplace at the clinics labeled “Recommended” though the course is Required
DQ-07	Manual duplicate accounts; one compliance cycle recorded on the duplicate
DQ-08	Test accounts with completions
DQ-09	Courses with no owner; one not reviewed since 2019
DQ-10	Agency accounts with no agency ID
PI-01	Aug 2024 scorecard built from 4-day-old data after an HR feed outage
PI-02	Mar 2025 Community North numerator double-counts March completions
PI-03	Oct 2025 Flagship Hospital includes agency accounts
AX-01 to 03	BI access: two terminated managers never revoked, one unit manager with system scope, one aggregate-only analyst with person-level detail

7. Known limits

- Benchmarks are national survey averages. NSI surveys acute-care hospitals only, so clinic and home-health rates are extrapolated.
- 2023 and 2025 wages are scaled from one 2024 benchmark, so treat them as approximate.
- Only RN hours are simulated at the unit-day level.
- No patient, clinical or quality outcomes, no demographics, no pay by person.
- Names come from short lists of common names, so different people share names on purpose.

- The published scorecard uses one deliberately naive definition (LMS status “completed” over all enrollments). Case study A replaces it.

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Personal project. All data is synthetic or public, and the company is fictional. Views are my own and not affiliated with or endorsed by any current or former employer.